

Optimization of energy communities in the Italian incentive system

Marta Stentati, Simone Paoletti, Antonio Vicino

Dipartimento di Ingegneria dell'Informazione e Scienze Matematiche

Università di Siena, Siena, Italy

marta.stentati@student.unisi.it, {paoletti, vicino}@diism.unisi.it

Abstract—In this paper we address the optimal operation of an energy community, which receives an incentive for the self-consumption realized at the community level in each time period. This incentive scheme mimics the one adopted for renewable energy communities in Italy since 2020. The operational problem is formulated as the maximization of the social welfare of the community over a given time horizon. The social welfare includes the incentive. Each entity of the community can encompass loads, generators, and battery energy storage systems. The optimization problem computes the battery charging/discharging policies, and the set points of flexible loads and controllable generators. The resulting problem formulation is non-convex, due to the presence of complementarity constraints. This issue is tackled by deriving an equivalent mixed integer linear programming formulation. A toy example and an application with real consumption, generation, and price data, are reported to illustrate the proposed approach.

Index Terms—Energy community, social welfare, non-convex problem, mixed integer optimization.

I. INTRODUCTION

Energy communities are aggregations of entities, that exchange energy among themselves through the public grid. The single entities may engage in generation (especially from renewable energy sources), consumption, energy storage, as well as provide energy services to other community members, or to the grid. Their participation in the energy community is voluntary and open. The primary purpose of energy communities is to provide environmental, economic, and social benefits to its members, or to the local areas where they operate [1]. Several studies have shown that aggregation is actually capable of generating benefits of different types [2].

While national or regional laws regulate the constitution of energy communities and their incentivization [1], an energy community may typically decide its own internal operation mechanisms. Energy volumes to be exchanged within a community can be obtained together with energy prices by clearing a suitably designed local energy market. This can be organized either in a centralized, or peer-to-peer fashion (see [3] for a comprehensive review). The operation mechanism of a community may also include a strategy to redistribute the community benefits among its participants [4].

This work was supported in part by the Italian Ministry for Research in the framework of the 2017 Program for Research Projects of National Interest (PRIN) under Grant 2017YKXYXJ.

In this paper, we do not assume that the energy community organizes a local energy market. Instead, we consider the energy community operating under an incentive scheme based on the self-consumption realized at the community level. This scheme has been adopted in Italy for renewable energy communities since 2020 [5]. Community self-consumption is defined as “the minimum between the energy fed into the network and the energy consumed by the community members” in a given time period, typically set to 1 hour. In this framework, the entities of a community may still operate in an uncoordinated fashion (i.e. each one acting individually), and then share the incentive resulting from the “incidental” self-consumption. However, if the entities optimally coordinate their energy flows, enhanced benefits are expected. Optimization of renewable energy communities under the Italian regulation is addressed by few works in the literature. In [6] and [7], optimization is included in methodologies aimed at evaluating business models and investments in active devices (renewable generators, energy storage systems) for the community. Optimal scheduling of a battery energy storage system (BESS) deployed within a small residential renewable energy community, is addressed in [8]. The contribution of this paper moves along a similar direction, but the model proposed here may also exploit the flexibility offered by demand and generation. Operation of the energy community is formulated as a non-convex optimization problem, whose objective is to maximize the social welfare (which includes the incentive) over a given time horizon. Non-convexity stems from complementarity constraints imposing no simultaneous export to, and import from the grid for each entity. To tackle this issue, an equivalent mixed integer linear program (MILP) is derived. A toy example, and an application with real consumption, generation, and price data, show that optimization increases the economic benefits for the community, compared to the uncoordinated solution.

The paper is organized as follows. Section II describes the original non-convex problem formulation, and the derivation of the equivalent MILP. Numerical results are illustrated in Section III. Finally, Section IV draws some conclusions.

II. PROBLEM FORMULATION

We consider an energy community composed of U entities. The entities are represented by grid-connected microgrids. To

TABLE I
NOMENCLATURE

Constants and sets

ΔT	duration of a time period [h]
U	number of entities forming the community
T	number of time periods
$\mathcal{U}$	set of all the entities, with $\mathcal{U} = \{1, \dots, U\}$
$\mathcal{T}$	set of the time indexes, with $\mathcal{T} = \{1, \dots, T\}$

Parameters

$I_{u,t}^{min}$	lower bound to energy demand of entity u at time t [kWh]
$I_{u,t}^{nom}$	nominal energy demand of entity u at time t [kWh]
$I_{u,t}^{max}$	upper bound to energy demand of entity u at time t [kWh]
$E_{u,t}^{min}$	lower bound to energy production of entity u at time t [kWh]
$E_{u,t}^{nom}$	nominal energy production of entity u at time t [kWh]
$E_{u,t}^{max}$	upper bound to energy production of entity u at time t [kWh]
S_u^{min}	lower bound to battery capacity of entity u [kWh]
S_u^{max}	upper bound to battery capacity of entity u [kWh]
S_u^{ini}	battery capacity of entity u at time $t = 0$ [kWh]
S_u^{end}	battery capacity of entity u at time $t = T$ [kWh]
P_u^{dis}	maximum discharging power of battery of entity u [kW]
P_u^{cha}	maximum charging power of battery of entity u [kW]
η_u^{dis}	discharging efficiency of battery of entity u
η_u^{cha}	charging efficiency of battery of entity u
$E_{u,t}^{cap}$	upper bound to power export of entity u to the grid at time t [kW]
$I_{u,t}^{cap}$	upper bound to power import of entity u from the grid at time t [kW]

$\pi_{u,t}^{egr}$	unit price of energy exported to the grid by entity u at time t [€/kWh]
$\pi_{u,t}^{igr}$	unit price of energy imported from the grid by entity u at time t [€/kWh]
$\pi_{u,t}^{fle}$	unitary cost for activating energy demand flexibility of entity u at time t [€/kWh]
$\pi_{u,t}^{reg}$	unitary cost for regulating energy production of entity u at time t [€/kWh]
$\pi_{u,t}^{sto}$	unitary cost for usage of energy storage of entity u at time t [€/kWh]
π_t^{inc}	unit incentive for community self-consumption at time t [€/kWh]

Variables

$i_{u,t}$	energy demand of entity u at time t [kWh]
$e_{u,t}$	energy production of entity u at time t [kWh]
$i_{u,t}^{gr}$	energy imported from the grid by entity u at time t [kWh]
$e_{u,t}^{gr}$	energy exported to the grid by entity u at time t [kWh]
$n_{u,t}^{gr}$	net energy exchange between entity u and the grid at time t [kWh]
$\delta_{u,t}$	binary variable expressing the sign of $n_{u,t}^{gr}$
$p_{u,t}^{dis}$	discharging power of battery of entity u at time t [kW]
$p_{u,t}^{cha}$	charging power of battery of entity u at time t [kW]
$s_{u,t}$	battery capacity of entity u at time t [kWh]
$\zeta_t, \epsilon_{u,t}^n$	auxiliary variables
$\epsilon_{u,t}^e, \epsilon_{u,t}^i$	

ease the notation, we assume that each entity can encompass a load, a generator, and/or a BESS. However, the model described hereafter can be straightforward extended to the case of entities including multiple loads, generators, and battery energy storage systems.

A. Non-convex formulation

For the generator of entity u , the energy production $e_{u,t}$ at time t has a nominal value $E_{u,t}^{nom}$, and it can be regulated within the interval defined by

$$E_{u,t}^{min} \leq e_{u,t} \leq E_{u,t}^{max}. \quad (1)$$

Similarly, for the load of entity u , the energy demand $i_{u,t}$ at time t has a nominal value $I_{u,t}^{nom}$. Still, it can be scheduled within a flexibility interval defined by

$$I_{u,t}^{min} \leq i_{u,t} \leq I_{u,t}^{max}. \quad (2)$$

Remark 1: The description above is general enough to model several types of generators/loads. For instance, a photovoltaic (PV) generator with curtailment can be modelled with $E_{u,t}^{min} = 0$, $E_{u,t}^{nom} = E_{u,t}^{max} > 0$, while a steerable generator is described by $E_{u,t}^{min} = E_{u,t}^{nom} = 0$, $E_{u,t}^{max} > 0$. A sheddable load is characterized by $I_{u,t}^{min} = 0$, $I_{u,t}^{nom} = I_{u,t}^{max} > 0$. $\square$

Additional constraints can be added to model specific features of generators and loads. For instance, the typical energy

payback effect of flexible domestic customers [9] can be taken into account by imposing

$$\sum_{t \in \mathcal{T}} i_{u',t} = \sum_{t \in \mathcal{T}} I_{u',t}^{nom} \quad (3)$$

for all $u' \in \mathcal{D}$, where $\mathcal{D} \subseteq \mathcal{U}$ is the set of entities with domestic loads.

For the BESS of entity u , the charging power $p_{u,t}^{cha}$ and discharging power $p_{u,t}^{dis}$ at time t are bounded as follows:

$$0 \leq p_{u,t}^{cha} \leq P_u^{cha} \quad (4)$$

$$0 \leq p_{u,t}^{dis} \leq P_u^{dis}. \quad (5)$$

Also the battery capacity $s_{u,t}$ at time t is bounded:

$$S_u^{min} \leq s_{u,t} \leq S_u^{max}. \quad (6)$$

Its dynamics over the time horizon $\mathcal{T}$ covered by the optimization, is given by the difference equation:

$$s_{u,t} = s_{u,t-1} + \eta_u^{cha} p_{u,t}^{cha} \Delta T - \frac{1}{\eta_u^{dis}} p_{u,t}^{dis} \Delta T, \quad (7)$$

with the initial and terminal conditions:

$$s_{u,0} = S_u^{ini} \quad (8)$$

$$s_{u,T} = S_u^{end}. \quad (9)$$

Remark 2: Notice that, to avoid simultaneous charging and discharging, one should in principle consider the complementarity constraint $p_{u,t}^{cha} p_{u,t}^{dis} = 0$. However, this constraint can be

omitted, being automatically satisfied at the optimum of the optimization problem presented in the following. $\square$

Regarding energy exchanges of entity u with the grid, these are bounded as follows at time t :

$$0 \leq e_{u,t}^{gri} \leq E_{u,t}^{cap} \Delta_T \quad (10)$$

$$0 \leq i_{u,t}^{gri} \leq I_{u,t}^{cap} \Delta_T. \quad (11)$$

Notice that the right-hand side inequalities of (10) and (11) reflect the contracted power for export and import at the connection of entity u with the grid. It is stressed that, at time t , entity u either exports energy to, or imports energy from the grid. This implies that the two quantities $e_{u,t}^{gri}$ and $i_{u,t}^{gri}$ cannot be simultaneously non-zero:

$$e_{u,t}^{gri} i_{u,t}^{gri} = 0. \quad (12)$$

Finally, the energy balance for entity u at time t , is given by the equation:

$$e_{u,t}^{gri} - i_{u,t}^{gri} - e_{u,t} + i_{u,t} - p_{u,t}^{dis} \Delta_T + p_{u,t}^{cha} \Delta_T = 0, \quad (13)$$

taking into account the net energy exchange with the grid, the energy produced and demanded by the entity, and the energy either supplied or absorbed by the BESS.

Equations (1)-(13), $\forall u \in \mathcal{U}$, $\forall u' \in \mathcal{D}$, $\forall t \in \mathcal{T}$, are the constraints of the optimization problem formulated for the operation of the energy community over the time horizon $\mathcal{T}$. The objective function (to be maximized) is the social welfare of the community, expressed by:

$$\begin{aligned} & \sum_{t \in \mathcal{T}} \sum_{u \in \mathcal{U}} [\pi_{u,t}^{egr} e_{u,t}^{gri} - \pi_{u,t}^{igr} i_{u,t}^{gri} \\ & - \pi_{u,t}^{reg} |e_{u,t} - E_{u,t}^{nom}| - \pi_{u,t}^{fle} |i_{u,t} - I_{u,t}^{nom}| \\ & - \pi_{u,t}^{sto} \Delta_T (\eta_{u,t}^{cha} p_{u,t}^{cha} + \frac{1}{\eta_{u,t}^{dis}} p_{u,t}^{dis})] \\ & + \sum_{t \in \mathcal{T}} \pi_t^{inc} \min \{ \sum_{u \in \mathcal{U}} e_{u,t}^{gri}, \sum_{u \in \mathcal{U}} i_{u,t}^{gri} \}. \end{aligned} \quad (14)$$

Equation (14) includes the revenues for selling energy to the grid, the costs for buying energy from the grid, the costs of regulating energy production away from the nominal value, the costs of activating flexibility of energy demand, the costs of battery usage, as well as the incentives received for self-consumption at the community level. Notice that the self-consumption at time t is expressed by

$$\min \{ \sum_{u \in \mathcal{U}} e_{u,t}^{gri}, \sum_{u \in \mathcal{U}} i_{u,t}^{gri} \}, \quad (15)$$

where $\sum_{u \in \mathcal{U}} e_{u,t}^{gri}$ is the total energy exported by the entities to the grid, and $\sum_{u \in \mathcal{U}} i_{u,t}^{gri}$ is the total energy imported by the entities from the grid.

Unfortunately, the formulated problem is non-convex, due to the complementarity constraints (12). To circumvent this issue, an equivalent MILP formulation is derived next.

B. Equivalent MILP formulation

To derive a MILP which is equivalent to the optimization problem formulated in Section II-A, for entity u at time t we introduce a variable $n_{u,t}^{gri}$, representing the net energy exchange with the grid. In other words,

$$n_{u,t}^{gri} \triangleq e_{u,t}^{gri} - i_{u,t}^{gri}. \quad (16)$$

Using (16), the complementarity constraint (12) can be omitted, and the energy balance (13) becomes:

$$n_{u,t}^{gri} - e_{u,t} + i_{u,t} - p_{u,t}^{dis} \Delta_T + p_{u,t}^{cha} \Delta_T = 0. \quad (17)$$

Moreover, constraints (10) and (11) are replaced with:

$$n_{u,t}^{gri} \leq E_{u,t}^{cap} \Delta_T \quad (18)$$

$$-n_{u,t}^{gri} \leq I_{u,t}^{cap} \Delta_T. \quad (19)$$

To formulate the objective function (14) using the new variables $n_{u,t}^{gri}$ in place of $e_{u,t}^{gri}$ and $i_{u,t}^{gri}$, we need the sign of $n_{u,t}^{gri}$. To this purpose, we introduce binary variables $\delta_{u,t}$ such that:

$$\delta_{u,t} = \begin{cases} 1 & \text{if } n_{u,t}^{gri} \geq 0 \\ 0 & \text{otherwise.} \end{cases} \quad (20)$$

In the MILP, the relation (20) is enforced through the constraints:

$$n_{u,t}^{gri} \leq M_\delta \delta_{u,t} \quad (21)$$

$$-n_{u,t}^{gri} \leq M_\delta (1 - \delta_{u,t}), \quad (22)$$

where $M_\delta > 0$ is a big-M parameter. Indeed, if $\delta_{u,t} = 1$, we have $n_{u,t}^{gri} \geq 0$ from (22), while (21) is not binding (provided that M_δ is chosen large enough). Vice versa, if $\delta_{u,t} = 0$, we have $n_{u,t}^{gri} \leq 0$ from (21), while (22) is not binding.

Using $n_{u,t}^{gri}$ and $\delta_{u,t}$, the terms $e_{u,t}^{gri}$ and $i_{u,t}^{gri}$ in (14) are replaced as follows:

$$e_{u,t}^{gri} = n_{u,t}^{gri} \delta_{u,t} \quad (23)$$

$$i_{u,t}^{gri} = n_{u,t}^{gri} (\delta_{u,t} - 1). \quad (24)$$

However, this introduces bilinear terms $n_{u,t}^{gri} \delta_{u,t}$ in the objective function. As is standard practice, these bilinear terms can be linearized by replacing them with auxiliary variables $\varepsilon_{u,t}^n$, and adding the constraints:

$$0 \leq \varepsilon_{u,t}^n \leq M_\varepsilon \delta_{u,t} \quad (25)$$

$$0 \leq \varepsilon_{u,t}^n - n_{u,t}^{gri} \leq M_\varepsilon (1 - \delta_{u,t}), \quad (26)$$

where $M_\varepsilon > 0$ is a big-M parameter.

Notice that, at this point, the self-consumption (15), is expressed as:

$$\min \{ \sum_{u \in \mathcal{U}} \varepsilon_{u,t}^n, \sum_{u \in \mathcal{U}} (\varepsilon_{u,t}^n - n_{u,t}^{gri}) \}. \quad (27)$$

To get rid of the min operator, we replace (27) with a new variable ζ_t , defined by the constraints:

$$\zeta_t \leq \sum_{u \in \mathcal{U}} \varepsilon_{u,t}^n \quad (28)$$

$$\zeta_t \leq \sum_{u \in \mathcal{U}} (\varepsilon_{u,t}^n - n_{u,t}^{gri}). \quad (29)$$

It turns out that, at the optimum of the MILP to be formulated next, ζ_t equals (27).

To complete the MILP formulation, we have finally to get rid of the absolute values in (14). This is done by replacing $|e_{u,t} - E_{u,t}^{nom}|$ and $|i_{u,t} - I_{u,t}^{nom}|$ with new variables $\varepsilon_{u,t}^e$ and $\varepsilon_{u,t}^i$, respectively, and adding the constraints:

$$e_{u,t} - E_{u,t}^{nom} \leq \varepsilon_{u,t}^e \quad (30)$$

$$E_{u,t}^{nom} - e_{u,t} \leq \varepsilon_{u,t}^e \quad (31)$$

$$i_{u,t} - I_{u,t}^{nom} \leq \varepsilon_{u,t}^i \quad (32)$$

$$I_{u,t}^{nom} - i_{u,t} \leq \varepsilon_{u,t}^i \quad (33)$$

Also in this case, at the optimum of the MILP we will have $\varepsilon_{u,t}^e = |e_{u,t} - E_{u,t}^{nom}|$ and $\varepsilon_{u,t}^i = |i_{u,t} - I_{u,t}^{nom}|$.

With all the replacements described above, the objective function (14) is reformulated as follows:

$$\begin{aligned} & \sum_{t \in \mathcal{T}} \sum_{u \in \mathcal{U}} [\pi_{u,t}^{egr} \varepsilon_{u,t}^n - \pi_{u,t}^{igr} (\varepsilon_{u,t}^n - n_{u,t}^{gr}) \\ & - \pi_{u,t}^{reg} \varepsilon_{u,t}^e - \pi_{u,t}^{fle} \varepsilon_{u,t}^i \\ & - \pi_{u,t}^{sto} \Delta_T (\eta_u^{cha} p_{u,t}^{cha} + \frac{1}{\eta_u^{dis}} p_{u,t}^{dis})] \\ & + \sum_{t \in \mathcal{T}} \pi_t^{inc} \zeta_t. \end{aligned} \quad (34)$$

The resulting MILP is the maximization of (34), subject to the constraints (1)-(9), (17)-(19), (21)-(22), (25)-(26), (28)-(33), $\forall u \in \mathcal{U}$, $\forall u' \in \mathcal{D}$, $\forall t \in \mathcal{T}$.

Remark 3: The formulated MILP requires $U \times T$ binary variables. This number can be significantly reduced in some cases, e.g. taking into account the characteristics of the entities. For instance, if entity u is a pure load (resp. generator), its binary variables $\delta_{u,t}$ can be directly set to 0 (resp. 1). $\square$

III. NUMERICAL RESULTS

In this section we present both a toy example, and an application with real consumption, generation, and price data.

A. Toy example

The aim of this simple example is to show that, in the considered setting of flexible demand and generation, maximizing the incentive may not lead to the most cost-effective solution for the community. Since we consider $\mathcal{T} = \{1\}$, i.e. a single time period, we drop the dependence on t to simplify the notation. The energy community is composed of three entities. The first entity is a sheddable load with $I_1^{min} = 0$, $I_1^{nom} = I_1^{max} = 5$ kWh, and $\pi_1^{fle} = 0.2$ €/kWh. Also the second entity is a sheddable load, with $I_2^{min} = 0$, $I_2^{nom} = I_2^{max} = 3$ kWh, and $\pi_2^{fle} = 0.4$ €/kWh. The third entity is a steerable generator with $E_3^{min} = E_3^{nom} = 0$, $E_3^{max} = 4$ kWh, and $\pi_3^{reg} = 0.05$ €/kWh. Grid prices are $\pi^{igr} = 0.3$ €/kWh and $\pi^{egr} = 0.1$ €/kWh. The unit incentive is set to $\pi^{inc} = 0.08$ €/kWh.

The maximum possible self-consumption is 4 kWh. This is achieved with $e_3^{gr} = 4$ kWh, and different feasible combinations of i_1^{gr} and i_2^{gr} summing up to 4 kWh. Among all these combinations, the less costly is the one with $i_1^{gr} = 1$ kWh

and $i_2^{gr} = 3$ kWh, leading to overall costs for the community equal to 1.48 €, net of the incentive of 0.32 €.

On the other hand, if the social welfare maximization is performed, the optimal solution is $i_1^{gr} = 0$, $i_2^{gr} = 3$ kWh, and $e_3^{gr} = 4$ kWh. The resulting self-consumption is 3 kWh, for a reduced incentive of 0.24 €. Nevertheless, the overall costs for the community improve to 1.46 €.

B. Real application

Load consumption data of 50 consumers connected to the same distribution network are available. Data cover the period from April 7, 2008 to October 23, 2008, for a total of $D = 200$ days. A normalized profile of PV generation is also available for the same time span. These data are used to simulate an energy community composed of $U = 51$ entities. Entity 1 is a PV generator equipped with a BESS. The PV generator has a nominal power of 450 kWp. Its generation profile $E_{1,t}^{nom}$ for day d is obtained scaling the normalized generation profile in that day of the data set by the nominal power. We assume that curtailment is not possible, hence we set $E_{1,t}^{max} = E_{1,t}^{min} = E_{1,t}^{nom}$. The maximum capacity of the BESS is $S_1^{max} = 600$ kWh, while full discharge is assumed to be possible ($S_1^{min} = 0$). The maximum charging and discharging power are set to $p_{1,t}^{cha} = p_{1,t}^{dis} = S_1^{max} / \Delta_T$. Charging and discharging efficiencies η_1^{cha} and η_1^{dis} are both equal to 0.95. The BESS is required to be fully discharged at the beginning and end of the day ($S_1^{ini} = S_1^{end} = 0$). For entity 1, energy import from the grid is not possible to avoid arbitrage ($I_{1,t}^{cap} = 0$), while the value of $E_{1,t}^{cap}$ is chosen greater than the nominal PV power. Entities numbered from 2 to 51 are consumers. For consumer u , the nominal profile $I_{u,t}^{nom}$ for day d is the load profile in that day of the data set. To simulate different levels of flexibility, lower and upper bounds are set to $I_{u,t}^{min} = (1 - \alpha) I_{u,t}^{nom}$ and $I_{u,t}^{max} = (1 + \alpha) I_{u,t}^{nom}$, where $\alpha = 0, 0.1$, and 0.25 in different tests. Moreover, $E_{u,t}^{cap} = 0$, since entity u does not export energy to the grid, while $I_{u,t}^{cap}$ is chosen greater than the contracted power. Prices of energy exported to the grid are those defined for the *Ritiro Dedicato* by GSE [10], while prices of energy imported from the grid are those defined for the *Servizio di Maggior Tutela* by ARERA [11]. In both cases, prices are those defined from April to October 2019. Hence, for consistency with the load consumption data, price profiles are suitably reconstructed using calendar days of 2008. For instance, the price valid for Sundays and holidays in April 2019, is applied to Sundays and holidays in April 2008. The unit incentive for community self-consumption is equal to 0.12 €/kWh for all time periods (the value currently set in Italy). The unitary cost of BESS usage is 0.01 €/kWh, while no cost for activating demand flexibility is considered.

The proposed optimization problem is solved for each of the D days of the data set. The number of time periods per day is $T = 24$, being $\Delta_T = 1$ h. Figure 1 shows the data and the results for a day of the data set in the case $\alpha = 0.25$. It can be noticed that from 10AM to 5PM the upward flexibility is exploited to maximize the amount

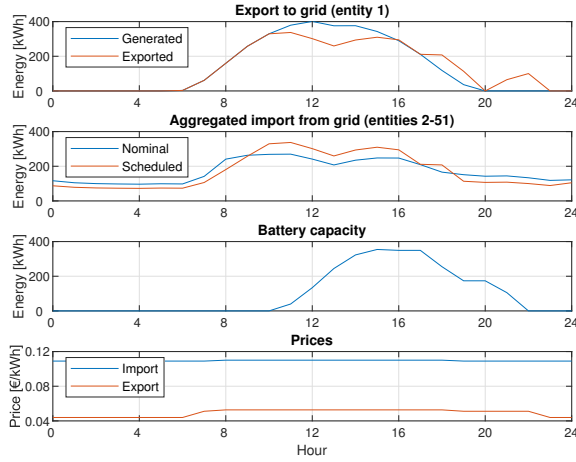


Fig. 1. Data and results for day 30, case $\alpha = 0.25$.

of PV generation self-consumed instantaneously. The excess of generated energy is stored in the BESS between 11AM and 3PM. Then, the BESS is progressively discharged from 6PM to 10PM, thus increasing community self-consumption at 6PM and 7PM (when PV generation is low), and enabling it at 9PM and 10PM (when PV generation is absent). The BESS is completely discharged by 10PM to avoid selling the exported energy at a lower price starting from 11PM. Notice that downward flexibility is used before 10AM and after 6PM to satisfy the energy payback constraints (3).

We consider three different scenarios:

- No community (NC): There is no community, and therefore no incentive. Each entity optimizes its own social welfare. This corresponds to solving the MILP without the incentive term in the objective function.
- Community uncoordinated (CU): As in scenario NC, each entity optimizes its own social welfare. The incentive is then computed *a posteriori*, and added to the optimal value of the objective function.
- Community coordinated (CC): The social welfare is optimized for the whole community by solving the MILP with the incentive term included in the objective function.

Table II shows the average net daily cost for the aggregated entities in the three scenarios for different values of α . In scenarios NC and CU the BESS is never used. Indeed, export prices are higher during daylight hours, when PV generation is active, and therefore entity 1 has no convenience in shifting the energy export. For scenario NC, the slight improvement of the average net daily cost with α is determined by shifting

TABLE II
AVERAGE NET DAILY COSTS FOR THE AGGREGATED ENTITIES [€]

α	NC	CU	CC
0	348.15	111.10	82.23
0.1	347.59	115.71	75.26
0.25	346.75	129.25	67.62

consumption to time periods where import prices are cheaper. For scenario CU, the average net daily cost increases with α , instead. This is due to the fact that single consumers find it convenient to shift consumption to night hours (when energy is only up to 0.01 €/kWh cheaper), but this causes a reduction of community self-consumption (remunerated 0.12 €/kWh) during daylight hours. For $\alpha = 0$ (no flexibility), the improvement from scenario CU to scenario CC is due to the use of the BESS to store the surplus of generated energy, thus enabling community self-consumption also in night hours, when PV generation is not available. Further improvements are obtained in scenario CC by introducing demand flexibility at increasing levels. Indeed, combining the use of the BESS and 25% demand flexibility, the average net daily cost reduction with respect to the case of “incidental” self-consumption (scenario CU with $\alpha = 0$) amounts to about 40%.

IV. CONCLUSIONS

This paper addressed the optimal operation of an energy community, which receives an incentive for the self-consumption realized at the community level. The considered problem was formulated as a MILP, computing the BESS charging/discharging policies, and the set points of controllable loads and generators, to maximize the social welfare (which includes the incentive) over a given time horizon. Future work is concerned with introducing uncertainty on demand and generation, through a stochastic formulation of the problem at hand. Moreover, participation of the community in the provision of different services, such as demand response and charging of electric vehicles, will be considered.

REFERENCES

- [1] “Directive (EU) 2019/944 of the European Parliament and of the Council,” *Off. J. Eur. Union*, vol. L, no. 158, pp. 125–199, 2019.
- [2] V. Z. Gjorgievski, S. Cundeva, and G. E. Georghiou, “Social arrangements, technical designs and impacts of energy communities: A review,” *Renew. Energy*, vol. 169, pp. 1138–1156, 2021.
- [3] T. Sousa, T. Soares, P. Pinson, F. Moret, T. Baroche, and E. Sorin, “Peer-to-peer and community-based markets: A comprehensive review,” *Renew. Sust. Energy Rev.*, vol. 104, pp. 367–378, 2019.
- [4] B. Cornélusse, I. Savelli, S. Paoletti, A. Giannitrapani, and A. Vicino, “A community microgrid architecture with an internal local market,” *Appl. Energy*, vol. 242, pp. 547–560, 2019.
- [5] M. L. Di Silvestre, M. G. Ippolito, E. Riva Sanseverino, G. Sciumè, and A. Vasile, “Energy self-consumers and renewable energy communities in Italy: New actors of the electric power systems,” *Renew. Sust. Energy Rev.*, vol. 151, pp. 1–13, 2021.
- [6] M. Moncecchi, S. Meneghello, and M. Merlo, “A game theoretic approach for energy sharing in the Italian renewable energy communities,” *Appl. Sci.*, vol. 10, no. 22, pp. 1–25, 2020.
- [7] A. Cielo, P. Margiaria, P. Lazzaroni, I. Mariuzzo, and M. Repetto, “Renewable energy communities business models under the 2020 Italian regulation,” *J. Clean. Prod.*, vol. 316, pp. 1–11, 2021.
- [8] G. Talluri, G. M. Lozito, F. Grasso, C. Iturrino García, and A. Luchetta, “Optimal battery energy storage system scheduling within renewable energy communities,” *Energies*, vol. 14, no. 24, pp. 1–23, 2021.
- [9] A. Garulli, S. Paoletti, and A. Vicino, “Models and techniques for electric load forecasting in the presence of demand response,” *IEEE Trans. Control Syst. Technol.*, vol. 23, no. 3, pp. 1087–1097, 2015.
- [10] Gestore dei Mercati Energetici (GSE). [Online]. Available: <https://www.gse.it/en/>
- [11] Autorità di Regolazione per Energia, Reti e Ambiente (ARERA). [Online]. Available: <https://www.arera.it/it/inglese/index.htm>